

What is a meaning of finetuning and why we do it?

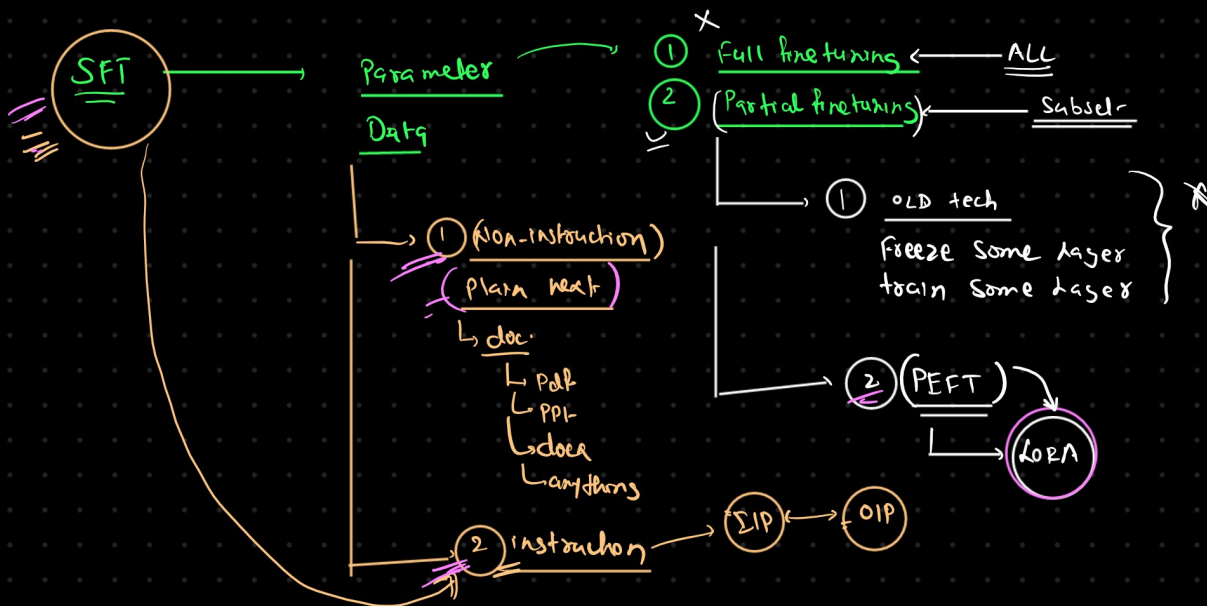
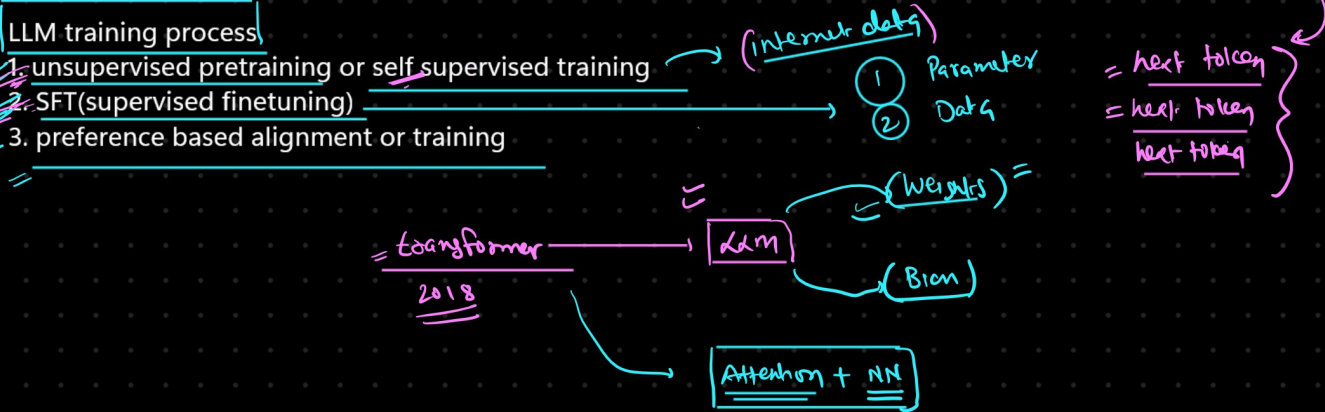
Fine-tuning means retraining an already trained (model) on your own specific dataset

LLM training process

1. unsupervised pretraining or self supervised training

2. SFT (supervised finetuning)

3. preference based alignment or training



Why it's called supervised fine-tuning (SFT)

• In pretraining → model predicts the next token from natural text itself (self-supervised, no human labels)

• In instruction fine-tuning → model predicts the next token from human-curated target answers (supervised learning)

Non instruction and Instruction finetuning?

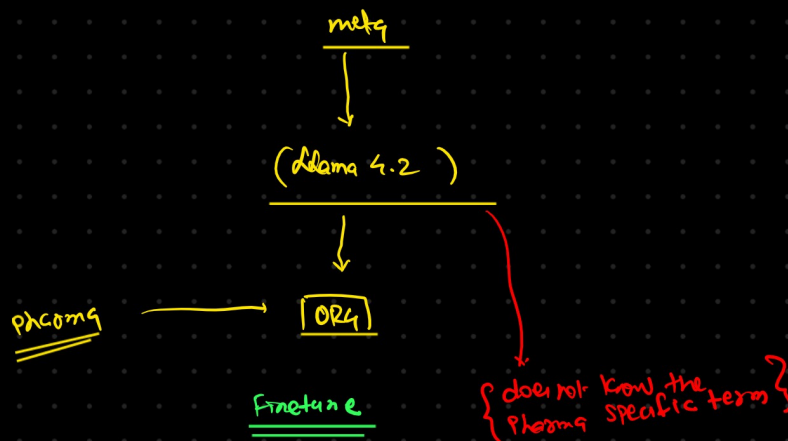
Fine-tuning on plain text (Non-instruction fine-tuning) means teaching the model your domain's language.

Purpose: To help the model learn the language, tone, and terminology specific to your domain.

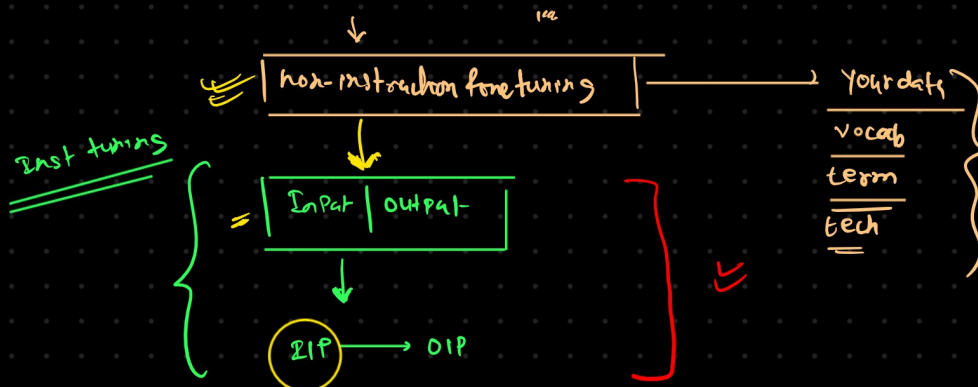
Data set example:

Metformin is one of the most widely prescribed oral antihyperglycemic agents.
Its primary mechanism of action involves the activation of AMP-activated protein kinase (AMPK),
a central metabolic regulator that promotes glucose uptake and fatty acid oxidation
while inhibiting hepatic gluconeogenesis.

Input (context)	Output (next token / continuation)
"Metformin is one of the most widely prescribed"	"oral"
"Metformin is one of the most widely prescribed oral"	"antihyperglycemic"
"Metformin is one of the most widely prescribed oral antihyperglycemic"	"agents."
...	...



Company specific data



Instruction fine-tuning means teaching the model how to follow user instructions.

Purpose: To train the model to understand instructions and generate well (structured), (relevant) answers.

Instruction-Following Dataset

```
{  
  "instruction": "Summarize the following paragraph.",  
  "input": "Metformin activates AMP-activated protein kinase (AMPK), which enhances glucose uptake...",  
  "output": "Metformin activates AMPK to improve glucose control and reduce gluconeogenesis."  
}
```

Without Input

```
{  
  "instruction": "List applications of AI in pharmaceutical research.",  
  "input": "",  
  "output": "Target identification, molecular docking, compound screening, and drug design optimization."  
}
```

```
{  
  "instruction": "Explain mechanism of Metformin.",  
  "input": "",  
  "output": "Metformin activates AMPK and enhances glucose uptake."  
}
```

- If instruction uses extra text (summarize / translate / rewrite) → use input
- If instruction is self-contained context or dialogue-style ya conversation data → keep input empty string ""

Instruction: "#### Input: "#### Response: " → AI Paq

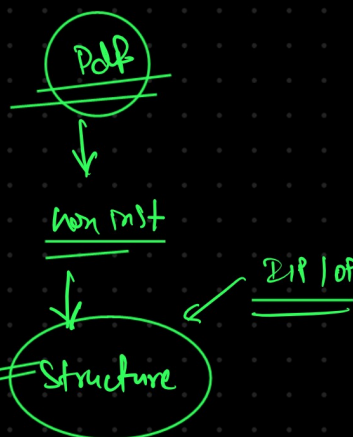
"Instruction: ", "Context: ", "Answer: " →

"USER: " / "ASSISTANT: "

"System: ", "User", "Assistant"

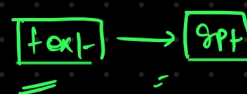
"context": "response"

```
{  
  "context": "I'm feeling worthless and can't sleep.",  
  "response": "You are not alone, therapy can help you regain balance."  
}
```



Now the Question is how to prepare the data for the instruction finetuning:

Method	Who Creates	Scale	Accuracy	Ideal Use
Manual	You / Annotators	Small	High	Education / General tasks
SME	Experts	Medium	Very High	Medical, Legal, Finance
LLM Synthetic	GPT / Claude	Large	Medium	Rapid domain datasets



prompt

{
"instruction": "Summarize the following paragraph.",
"input": "Metformin activates AMP-activated protein kinase (AMPK), which enhances glucose uptake...",
"output": "Metformin activates AMPK to improve glucose control and reduce gluconeogenesis."
}

During fine-tuning, the model is given a single continuous text (not three separate columns).

So, we need to combine all three parts of each row into one continuous training text.

A single training sample for the model would look like this:

Instruction: Summarize the following paragraph. ### Input: Metformin activates AMP-activated protein kinase (AMPK), which increases glucose uptake and reduces hepatic gluconeogenesis. ### Response: Metformin enhances glucose uptake and reduces glucose production by activating AMPK.

long
modeling

→ next token pool

Why instruction finetuning required

A base LLM (like LLaMA, Falcon, or GPT-NeoX) only learns text continuation.

This means it understands patterns such as:

"If this sentence is written, what word should come next?"

Example:

Input: Metformin is used for

Output: the treatment of type 2 diabetes mellitus.

So, the base model understands grammar and knowledge, but it doesn't understand instructions like:

"Explain this," "Summarize this," "Write in points," or "Translate this."

What it is

If you ask the base model:

"Write 3 points on Metformin mechanism."

it might respond with something like:

"Write 3 points on Metformin mechanism is a request..."

— meaning it treats your instruction as text, not as a command to follow.

So the conclusion is:

To teach the model the meaning and purpose of different prompts, we perform instruction fine-tuning.

Here, we train the model to understand that the user's prompt is an instruction, and the model must respond to it, not just continue the text.

The dataset format looks like this:

```
{"instruction": "Explain mechanism of Metformin.",  
"input": "",  
"output": "Metformin activates AMPK which enhances glucose uptake and inhibits gluconeogenesis."}
```

Result:

Now, the model doesn't just write — it actually understands

Now it can:

- Summarize information
- Answer questions
- Compare topics
- Explain code
- Maintain dialogues

This is the exact step followed by models like ChatGPT, Alpaca, Dolly, and Open-Orca.

